

A REVIEW OF MINIMIZATION TECHNIQUES FOR NONLINEAR FUNCTIONS¹

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INTRODUCTION

THE ADVENT OF THE DIGITAL COMPUTER and the desire to optimize systems have given a large impetus to the study of methods to obtain the minimum value of a nonlinear function. In mathematical terms we say that for a given function of n independent variables,

$$(1) \quad f(X) = f(x_1, x_2, \dots, x_n),$$

we wish to find that value of the independent variables, $X^* = [x_1^*, x_2^*, \dots, x_n^*]$ such that $f(X^*)$ is a minimum subject to the restrictions that

$$(2) \quad g_i(x_1^*, \dots, x_n^*) = 0 \quad i = 1, \dots, p < n,$$

$$g_i(x_1^*, \dots, x_n^*) \geq 0 \quad i = p + 1, \dots, m.$$

For an unconstrained function with continuous derivatives, a minimum occurs at that point where the partial derivatives of the function with respect to the independent variables are zero and its matrix of second derivatives is positive definite. A similar criteria can be stated for a constrained function by using Lagrange multipliers. We note that for a function with n independent variables, this approach requires the solution of at least n nonlinear equations and is, thus, not practical.

The purpose of this paper is to review the existing literature on minimization techniques. While it would be impossible to include all variations, the author has endeavored to include the basic methods and some of the more significant modifications. The emphasis will be on those methods which are primarily useful when a mathematical expression for the function can be obtained. The practical difficulties associated with the experimental and "on-line" use of these methods will not be investigated. To avoid confusion, all methods will be discussed in terms of locating a minimum. However, these methods can be applied to finding the maximum of a function with only minor changes.

A minimization procedure for a digital computer provides an algorithm by which the function is tested at a set of points. These points then provide information about the function and the location of its minimum. How the test points are chosen divides these procedures into two very general classes: sequential and nonsequential. In a sequential procedure the test points are determined by a fixed set of operations. The values of the independent variables are completely determined by the previous measurements. In the first part of this paper we shall discuss in some detail those methods which in some way depend on measurements of the slope of the function to determine the next test point. In Part II, we dis-

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cuss a few of the methods which have been developed to handle functions whose independent variables are constrained. We shall limit our discussion to those methods which convert the constraining problem into an equivalent unconstrained form. We note that there have been developed other methods for handling this constraint problem which we will not cover. The reader is referred to a recent comprehensive survey* of this problem by P. Wolfe [36]. In part III, two methods will be discussed which reduce the range of the independent variables a predetermined amount at each step in the iterative procedure. In a non-sequential procedure the points are picked essentially at random. This type of search procedure will be discussed further in Part IV.

In this paper, the author has omitted those methods which use a low order polynomial fitted to previously measured points as a means of extrapolating toward the minimum. A typical procedure is given by Lancaster [1]. These methods require a considerable amount of computation to obtain the polynomial fit. However, there is no real correlation between the minimum of the polynomial and the minimum of the original function. Thus it is the author's opinion that the time spent in computation could be better spent by using one of the simpler procedures to be discussed although more iterations may be required.

We should note that the methods to be discussed locate points where the derivatives with respect to the independent variables are zero. If the function has only one such point in a given region, the function will be called unimodal in that region, and the point will be an absolute minimum. If the function has more than one point, it will be called multimodal. In that case, the point must be tested to determine whether the point is a minimum, a saddle point or a maximum [2]. Even if a relative minimum has been obtained, the search procedure must be continued until all minima have been located if the absolute minimum is to be found. As yet no systematic method has been developed to handle multimodal functions.

PART I: GRADIENT METHODS

The largest body of material in the literature is concerned with what we shall define as "Gradient Methods" of minimization. These methods use measurements of the slope of the function as an indication of the direction toward the minimum. Intuitively this approach has considerable appeal. The first test point is picked arbitrarily and depends primarily on the programmer's subjective opinion as to the location of the minimum. To reduce computing time, it is obvious that this point should be picked as close as possible to the minimum. The remaining test points are determined by the iterative equation [2]:

$$(3) \quad X^{p+1} = X^p + h^p D^p$$

where X^p is an n -dimensional vector whose components are the independent variables $x_1, \dots, x_n$ at the p th iteration, h^p is a positive constant and D^p is an n -dimensional direction vector evaluated at the p th iteration. The vector D^p determines the direction to be taken from the p th point and the magnitude of

* The author wishes to thank the referee for bringing this survey to his attention.

$h^p D^p$ determines how large a step is made in that direction. The various gradient methods differ in their choice of the scale factor h^p and the direction vector D^p . In the following sections we shall discuss gradient methods in more detail under the assumption that the function to be minimized is unconstrained.

1.1. UNIVARIATE OR RELAXATION METHODS

Conceptually, the simplest method is

$$(4) \quad D^p = E_j^p$$

where

$$E_j = \begin{bmatrix} 0, 0 & \cdots & 0, 1, 0 & \cdots & 0 \\ & & j \text{ th row} & & \end{bmatrix}$$

In other words, we change only one coordinate at a time. These methods are called univariate or relaxation methods. In the Southwell relaxation method a coordinate is chosen for which $\partial f / \partial x_j$ has the largest value. A slight modification of this method is the Southwell-Synge method [3] in which we choose the coordinate for which

$$\left(\frac{\partial^2 f}{\partial x_j^2} \right) / 2 \frac{\partial^2 f}{\partial x_j^2}$$

is the largest. In both cases the constant h^p can be determined by forming a Taylor series about the point x^p . Thus,

$$(5) \quad f(x^{p+1}) = f(x^p) + h^p \left. \frac{\partial f}{\partial x_j} \right|_{x=x^p} + \frac{(h^p)^2}{2} \left. \frac{\partial^2 f}{\partial x_j^2} \right|_{x=x^p}$$

For $f(x^{p+1})$ to be a minimum, we see that

$$(6) \quad h^p = \frac{\left. \frac{\partial f}{\partial x_j} \right|_{x=x^p}}{\left. \frac{\partial^2 f}{\partial x_j^2} \right|_{x=x^p}}.$$

Since evaluating the second derivative along the given coordinate can be time consuming, an approximation to the optimum constant can be used. It should be noted that the second derivative must be positive. If the curve has a point of inflection, the iterative procedure may diverge away from the minimum point. A more complete discussion can be found in Booth [3].

Booth also asserts that these methods will in general take n times as much time as the steepest descent and other methods to be discussed in the next section. It is evident that making a simultaneous change in all independent variables will reduce the number of test points. However, the author is not convinced that the total amount of computer time necessary to obtain the minimum will be n times longer since the iterative operations are quite simple.

A variation [4] on this method tests three or four points along a coordinate and fits a low order polynomial to these points. After finding the minimum value

of the polynomial, that coordinate is held constant while the other variables of the function are similarly varied. As has been discussed, the increase in speed of convergence of the test points toward the minimum may not always compensate for the increased time necessary to evaluate the polynomial.

We should note that if the axis of the contours

$$(7) \quad f(x_1, \dots, x_n) = c$$

(c is an arbitrary constant) is inclined at 45 degrees to the coordinate axis, the univariate method may stop before reaching the minimum. As we can see in Fig. 1 for two-dimensions, point A is a minimum relative to the points B and C. Thus, the procedure will stop unless the step size is reduced by half. In that case the search will proceed to D and again the step size will have to be reduced. Thus, we see for such a function the step size is quite critical.

1.2. THE METHODS OF STEEPEST DESCENT, NEWTON, AND "OPTIMUM GRADIENT"

Methods which change all the independent variables at each iteration primarily use the following direction vector.

$$(8) \quad D^p = -B^{-1}\Delta^p$$

where

$$\Delta^p = \left[\left. \frac{\partial f}{\partial x_1} \right|_p, \left. \frac{\partial f}{\partial x_2} \right|_p, \dots, \left. \frac{\partial f}{\partial x_n} \right|_p \right],$$

and B is a positive definite $n \times n$ weighting matrix. The oldest and simplest weighting on the gradient vector Δ_p is

$$(9) \quad B = I$$

where I is the identity matrix. This was first described by Cauchy [5] in 1847 and is known as the method of steepest descent. It has been referred to by many

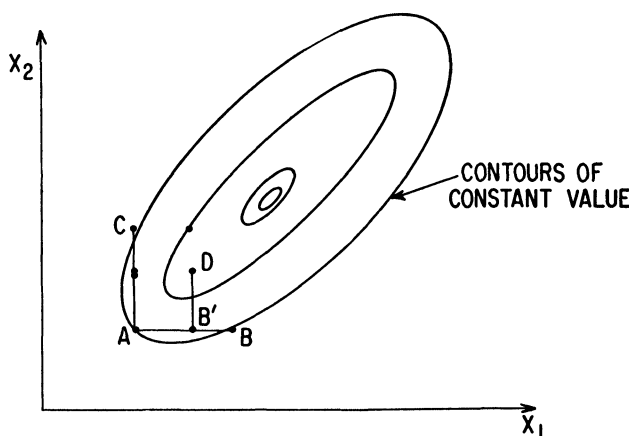


FIGURE 1. A UNIVARIATE SEARCH

authors since that time but has not been used as widely as might be expected because of its unpredictable rate of convergence toward the minimum. Curry [6] has noted that the convergence is not invariant under certain elementary transformations of the independent variables. It can be seen in Fig. 2 that as a result of changing the scale factor of one of the variables of the function

$$(10) \quad y = +(x_1 - 2)^2 + (x_2 - 2)^2$$

the gradient no longer points in the direction of the minimum.

In an effort to improve convergence, one can approximate $f(X^{p+1})$ by the first three terms of a generalized Taylor series. It can be shown that the optimum weighting is

$$(11) \quad B = L,$$

where L is the $n \times n$ matrix of second derivatives having components

$$(12) \quad l_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

This method is known as Newton's method. Crockett and Chernoff [7] have shown that the ratio of the largest to the smallest eigenvalue of L provides a measure of the convergence to a solution, and if B equals the matrix of second derivatives at the minimum then the convergence is as fast as possible. Only for a quadratic function can we obtain the best convergence. In general, the matrix of second derivatives varies at each test point. To reduce iteration time, the matrix is generally modified only after a preassigned number of iterations have occurred. Keeping the weighting matrix constant will increase the number of iterations. However, the over-all computer time may possibly be reduced since each iteration will be faster.

We note that B must be a positive definite matrix for the iterative process to proceed toward the minimum. Thus Newton's method can only be used in a region near a minimum unless the function is convex since L will not in general remain positive definite. For the initial test points which are far from the mini-

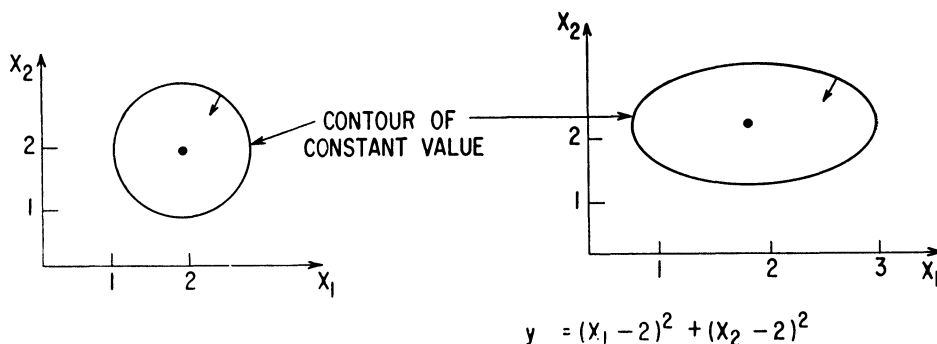


FIGURE 2. THE EFFECT OF A CHANGE IN SCALE ON THE DIRECTION OF THE GRADIENT

imum, the method of steepest descent or other method with a positive definite B should be used. The reader is referred to R. R. Brown [2] and W. C. Davidon [8] for a discussion of more general weighting matrixes. Davidon also discusses an iterative method for modifying the weighting matrix depending on the changes of gradient with changes of the independent variables.

From Eqs. (3) and (8), we see that

$$(13) \quad X^{p+1} = X^p - h^p B^{-1} \Delta^p$$

The size of the step in direction $B^{-1} \Delta^p$ is determined by the magnitude of

$$h^p B^{-1} \Delta^p.$$

It can be shown [6, 7] that the iterative process will eventually converge to the minimum if

$$(14) \quad J(X^{p+1}) < f(X^p).$$

Thus, a sufficient requirement on h^p is that the above inequality be satisfied. An elementary criterion for adjusting h^p uses the previous value if the inequality is satisfied and half the previous value if the inequality is not satisfied. We should note that in this case h^p has no way of increasing, and many iterations may be required to reach the minimum especially if the gradient is large at the initial test points. Although the inequality is satisfied, the gradient at the new point may be essentially in the opposite direction compared to the previous gradient. In that case, a large number of iterations may also be necessary for a very small increase in the direction of the minimum as shown in Fig. 3.

Booth [3] suggests using that value of h^p such that X^{p+1} is the intersection of the tangent to the function at the point X^p and the coordinate axis. From Fig. 4,

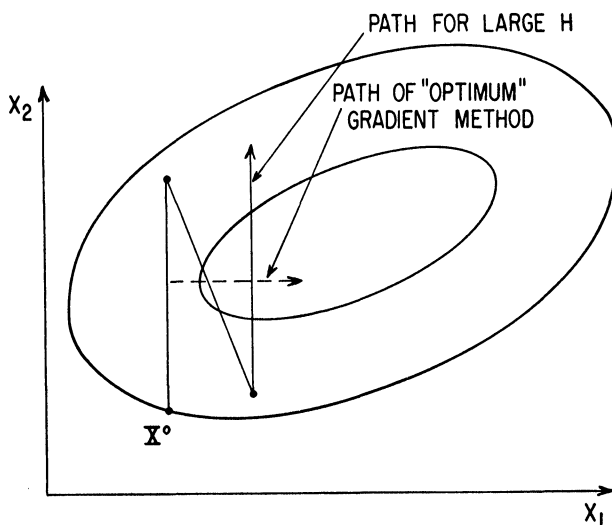


FIGURE 3. THE EFFECT OF h^p ON CONVERGENCE

we see that along the direction $B^{-1}\Delta^p$ the difference $X^{p+1} - X^p$ is given by

$$(15) \quad (X^{p+1} - X^p)' \cdot (B^{-1}\Delta^p) = f(X^p),$$

where $(X^{p+1} - X^p)'$ is the transpose of $(X^{p+1} - X^p)$. But

$$(16) \quad (X^{p+1} - X^p)' = h^p (B^{-1}\Delta^p).$$

Therefore,

$$(17) \quad h^p = \frac{f(X^p)}{(B^{-1}\Delta^p)' \cdot (B^{-1}\Delta^p)}$$

We note that when $B = I$, this is the Newton-Raphson method for n dimensions. It can also be seen that close to the minimum this value of h^p causes X^p to oscillate around the minimum. Thus, this value of h^p should be used only when the desired minimum has a value of zero.

It is clear from Fig. 3 that the optimum value of h^p is that which makes $f(X^{p+1})$ a minimum along the line $B^{-1}\Delta^p$. At that point

$$(18) \quad (\Delta^{p+1})' \cdot B^{-1}\Delta^p = 0,$$

where $(\Delta^{p+1})'$ is the transpose of Δ^{p+1} . Thus, the new direction will be 90 degrees to the previous direction. This choice of h^p is called the "optimum" gradient method and can be used with any weighting matrix, B . For a quadratic function the minimum will be obtained in one iteration using Newton's method and this "optimum" scale factor.

To obtain the exact optimum, one would require many iterations along the line $B^{-1}\Delta^p$. Therefore, several approximations have been developed. One approximation [9] depends on the angle, θ , between $B^{-1}\Delta^p$ and Δ^{p+1} . If $\theta > 90^\circ$, h^{p+1} is reduced and, if $\theta < 90^\circ$, h^{p+1} is increased. Another approximation [3] uses a small value of h so that three points are obtained along $B^{-1}\Delta$. A quadratic polynomial is passed through these points and the location of its minimum is selected as X^{p+1} . Of course, higher order extrapolations can be used, but the increased time necessary to evaluate the coefficients may not justify the increased accuracy.

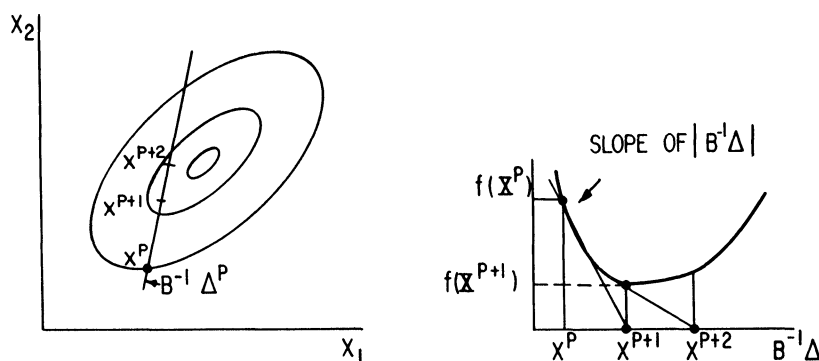


FIGURE 4. THE STEEPEST DESCENT ALONG A TANGENT

While the optimum gradient method is probably the most efficient for minimizing a generalized function, slow convergence is still a problem. One way of accelerating the convergence of the method is to systematically choose a value of h^p which is slightly over or slightly under the optimum. The results of numerical tests have shown [10] that this causes self-acceleration at random intervals.

Another acceleration technique suggested by Forsythe and Motzkin [11, 12] uses, after a certain number of iterations, the direction vector

$$(19) \quad D^p = D^{p-1}h^{p-1} + D^{p-2}h^{p-2}$$

This new direction vector in effect averages the previous directions as shown in Fig. 5. If the minimum in the directions D^{p-2} and D^{p-1} have been found exactly, we see that the tangent to the contour at X^{p-2} is parallel to D^{p-1} . For a quadratic surface we know from the theorem of parallel chords [13] that direction D^p passes through the absolute minimum. For nonquadratic surfaces, further iterations are necessary. The theorem is only valid for two independent dimensions. However, this method has been reported by Finkel [13] to be fairly successful for $n > 2$.

It is clear that other weightings and scale factors are possible for functions having special properties. The reader is referred to Refs. 3, 14, and 15 for more details.

Throughout this section, we have implicitly assumed that the first and second derivatives of $f(X)$ are analytically available. If this is not true, then finite difference approximations to the derivatives must be used. For completeness, we shall list below the necessary formulas as given by Brown [2]. Let

$$(20) \quad f_{\alpha\mu} = f(x_1^0 + \alpha h, x_2^0 + \mu h)$$

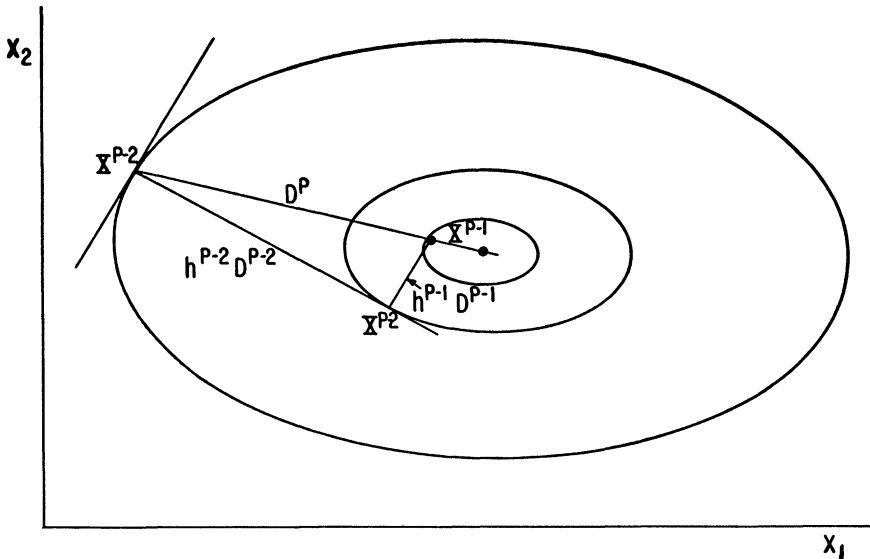


FIGURE 5. THE FORSYTHE ACCELERATION METHOD

Then, the approximations are:

$$(21) \quad \left. \frac{\partial f}{\partial x} \right|_{\substack{x=x_0 \\ y=y_0}} = \frac{1}{h} (f_{1,0} - f_{0,0});$$

$$(22) \quad \left. \frac{\partial f}{\partial y} \right|_{\substack{x=x_0 \\ y=y_0}} = \frac{1}{h} (f_{0,1} - f_{0,0});$$

$$(23) \quad \left. \frac{\partial^2 f}{\partial x^2} \right|_{\substack{x=x_0 \\ y=y_0}} = \frac{1}{h^2} (f_{-1,0} - 2f_{0,0} + f_{1,0});$$

$$(24) \quad \left. \frac{\partial^2 f}{\partial y^2} \right|_{\substack{x=x_0 \\ y=y_0}} = \frac{1}{h^2} (f_{0,-1} - 2f_{0,0} + f_{0,1});$$

$$(25) \quad \left. \frac{\partial^2 f}{\partial x \partial y} \right|_{\substack{x=x_0 \\ y=y_0}} = \frac{1}{h^2} (f_{0,0} - f_{0,1} - f_{1,0} + f_{1,1}).$$

A geometrical picture of the necessary points is given in Fig. 6. The errors associated with these formulas are given in Brown [2]. For only a slightly greater number of points, Brown has obtained quadratic approximations to the derivatives which have considerably less error.

1.3. DIRECT SEARCH

A search technique which combines some of the aspects of the univariate and steepest descent methods has been developed by Hooke and Jeeves [16] and Wood [17] and is called by them "Direct Search". A typical search pattern in

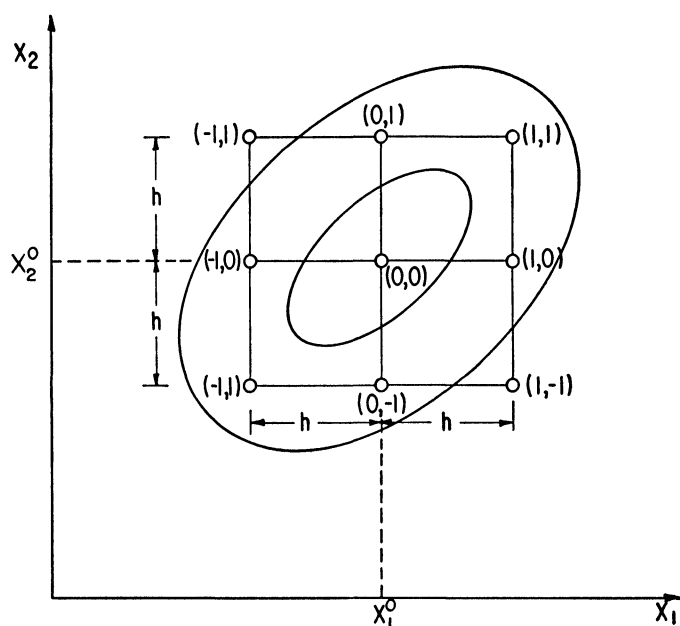


FIGURE 6. DERIVATIVE EVALUATION BY FINITE DIFFERENCING

two dimensions is shown in Fig. 7. After the first point has arbitrarily been picked, a univariate search is made around that point to determine the direction toward the minimum. Having individually changed each independent variable by a given increment, h , a point (in this case, point 3) is obtained which is less than any previous point. A "pattern move" is next made which consists of changing all independent variables an amount equal to the sum of the changes made in those variables during the previous two moves or searches. Thus, in our example, a move is made to point 4. A univariate search is made to determine the minimum value around this new point and the result is compared with the value obtained before the move. By making this search, the effective pattern move need not be always 45 degrees to the coordinate axis. If the result is less than the previous point, then another pattern move is made. However, if the result after any pattern move is larger, a univariate search is performed at the previous point to determine a new direction for a move. Eventually, this univariate search will be unable to locate a point which is less than the established point. In that case the step size of the search is reduced and another search is performed. This process of searches and pattern moves is continued until the step size of a search is reduced to predetermined minimum. Complete details on this method are given by Wood [17].

We see that the method combines the simplicity of the univariate search with

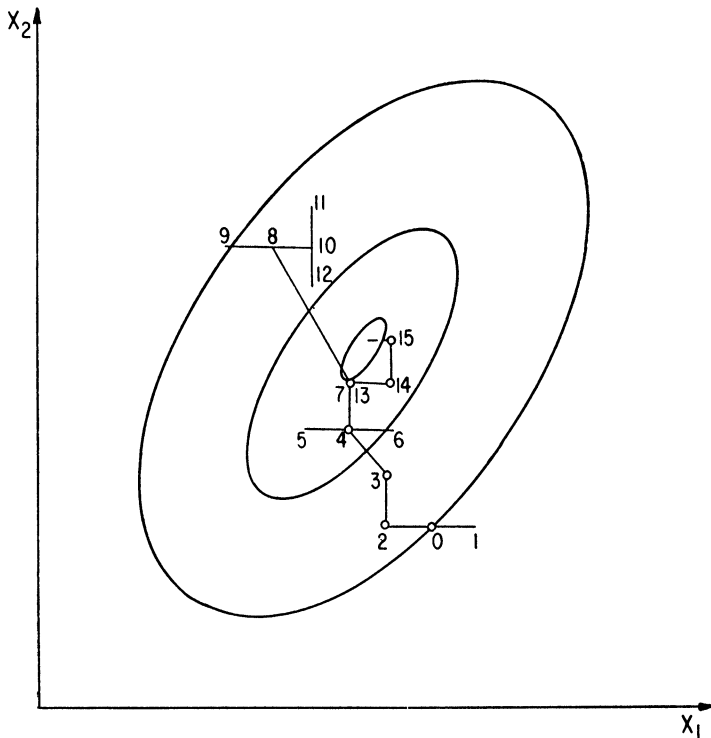


FIGURE 7. A TYPICAL "DIRECT" SEARCH

the larger steps which can be obtained from the steepest descent method. The more pattern moves we have in a given direction, the more confidence we have that the minimum lies in that direction and, therefore, the step size is increased. We should note that if the axis of the contours of equal value are inclined at 45 degrees to the coordinate axis, the difficulty shown in Fig. 1 may occur. This difficulty is not as serious as in the univariate method since large pattern moves can still be made.

PART II: GRADIENT METHODS FOR A FUNCTION WHOSE INDEPENDENT VARIABLES
ARE CONSTRAINED

In the previous part, we discussed the problem of minimizing a function without any constraints on the independent variables. We shall now consider a few methods for minimizing

$$(26) \quad f(X)$$

subject to the constraints that

$$(27) \quad g_i(X) = 0 \quad i = (1, 2, \dots, p \leq n),$$

$$(28) \quad g_i(X) \geq 0 \quad i = (p + 1, \dots, m).$$

Since the constraint

$$(29) \quad r(X) \leq 0$$

can be written as

$$(30) \quad g(X) = -r(X) \geq 0$$

we see that the minimization problem given by Eqs. (26), (27) and (28) is completely general.

If $f(X)$ and its constraints are linear, this problem can be handled by linear programming techniques. It is fairly clear that the solution will be on the boundary of the region determined by the constraints. The simplex method developed by Dantzig [18] makes use of this fact to obtain a solution in a finite number of iterations. Several other methods have been developed primarily to facilitate solution when there are a large number of constraints.

If $f(X)$ is non-linear, the minimum may no longer be on the boundary. Several extensions of linear programming techniques have been developed when the constraints are linear. Kelly's [19] "Cutting Plane" method and Rosen's [20, 21] "Gradient Projection" method are recent examples which seem to show considerable promise. These methods can also be used when the constraints are non-linear by using tangent plane approximations to the constraints.

A second approach to the problem has been to convert the constrained problem into one which is not constrained. This is particularly useful when both the function and its constraints are non-linear. In the following sections, the author will discuss in more detail this approach. For other techniques, the reader is referred to an extensive bibliography on linear and non-linear programming compiled by Riley and Gass [22].

2.1. DIRECT SEARCH

The direct search method provides a simple method of handling constraints. The actual value of $f(X)$ is replaced by a very large value whenever the constraints are not satisfied. This pseudo-value must be proportional to the amount the inequality is violated so that the search routine will force the test points into the region where the inequality can be satisfied. Thus, a typical sequence corresponding to Eqs. (26), (27), and (28) would be

$$(31) \quad \begin{aligned} SN(X) &= f(X) \begin{cases} g_i(X) = 0 & i = (1, 2, \dots, p \leq n) \\ g_i(X) \geq 0 & i = (p+1, \dots, m) \end{cases} \\ SN(X) &= 1 \times 10^{20} \left(\sum_{\alpha} |g_{\alpha}(X)| \right) + f(X) \end{aligned}$$

where $SN(X)$ is the value of the function used in the search and the sum, α , consists of those $g_i(X)$ which are not satisfied. $SN(X)$ is now an unconstrained function whose minimum is within or on the boundary of the region defined by the constraints.

2.2. LAGRANGIAN METHOD

A method which can be used with any of the gradient methods given in the previous sections is based on the classical Lagrangian function [23–28]. To utilize the Lagrangian function, we must convert the inequality constraint given in Eq. (28) into an equality constraint by introducing a non-negative “slack” variable, z_i . Thus

$$(32) \quad \begin{aligned} g_i(X) - z_i &= 0 & i &= (p+1, \dots, m) \\ z_i &\geq 0 \end{aligned}$$

We can now define in the usual manner the Lagrangian function,

$$(33) \quad \Phi(X, \lambda, Z) = f(X) - \sum_{i=1}^p \lambda_i g_i(X) - \sum_{i=p+1}^m \lambda_i [g_i(X) - z_i]$$

where λ_i is a non-negative variable independent of X . It can be shown [23] that a necessary and sufficient condition for X^* to be a solution of the minimization problem given in Eqs. (26), (27), and (28) is that $f(X^*)$ be concave and the constraint set be convex in the vicinity of X^* and that X^* , λ^* , and Z satisfy the following set of equations:

$$(34) \quad \frac{\partial \Phi}{\partial x_j^*} = \frac{\partial f(X^*)}{\partial x_j^*} - \sum_{i=1}^m \lambda_i^* \frac{\partial g_i(X^*)}{\partial x_j^*} = 0$$

$$(35) \quad \frac{\partial \Phi}{\partial \lambda_i^*} = -g_i(X^*) = 0 \quad i = 1, \dots, p;$$

$$(36) \quad \frac{\partial \Phi}{\partial \lambda_i^*} = -g_i(X^*) + z_i = 0 \quad i = p+1, \dots, m;$$

$$(37) \quad \begin{aligned} z_i &\geq 0, & i &= p+1, \dots, m \\ \lambda_i^* &\geq 0, & i &= 1, \dots, m, \end{aligned}$$

$$(38) \quad \lambda_i^* z_i = 0 \quad i = p+1, \dots, m.$$

We observe from Eqs. (36) and (38) that Z is completely determined by X^* and λ^* . It can also be shown [24] that

$$(39) \quad \Phi(X^*, \lambda, Z) \leq \Phi(X^*, \lambda^*, Z) \leq \Phi(X, \lambda^*, Z)$$

In other words, the Lagrangian function converts the constrained minimization problem into an unconstrained saddle point problem.

K. Arrow and H. Uzawa [29] have developed an iterative gradient procedure for locating this saddle point. Corresponding to Eq. (13) the iterative formulas are:

$$(40) \quad X^{p+1} = X^p - h^p B_1^{-1} \Delta_x^p$$

and

$$(41) \quad \lambda^{p+1} = \max [0, \lambda^p + h^p B_2^{-1} \Delta_\lambda^p]$$

where

$$(42) \quad \Delta_x^p = \left[\frac{\partial \Phi}{\partial x_1}, \dots, \frac{\partial \Phi}{\partial x_n} \right] \Big|_{x=X^p}$$

$$(43) \quad \Delta_\lambda^p = [-g_1(X^p), \dots, -g_m(X^p)]$$

h^p a constant and B_1 and B_2 are positive definite weighting matrices. The convergence of this method to the saddle point depends on the weighting matrices and the constant, h^p as was discussed in section 1.2.

2.3. BROWN'S METHOD

Eqs. (26), (27) and (32) define $n - p$ independent and m dependent variables. If it were possible to eliminate the dependent variables, the resultant unconstrained function could be minimized with respect to the independent variables. Brown [2] has developed a gradient method which solves these equations implicitly. Let us label the independent variables as $w_1, \dots, w_{n-p}$ and the dependent ones as $u_1, \dots, u_m$. Then, equivalent to Eqs. (26), (27), and (32), we wish to minimize

$$(44) \quad f(W, U) = f(w_1, \dots, w_{n-p}, u_1, \dots, u_m)$$

subject to the constraint that

$$(45) \quad g_i(W, U) = g_i(w_1, \dots, w_{n-p}, u_1, \dots, u_m) = 0 \quad (i = 1, \dots, m)$$

The derivative of $f(W, U)$ with respect to the independent variable w is

$$(46) \quad \Delta_c = F_w + F_u U_w,$$

where

$$F_w = \left[\frac{\partial f}{\partial w_1}, \dots, \frac{\partial f}{\partial w_{n-p}} \right] \quad F_u = \left[\frac{\partial f}{\partial u_1}, \dots, \frac{\partial f}{\partial u_m} \right]$$

and

$$U_w = \begin{bmatrix} \frac{\partial u_1}{\partial w_1}, \dots, \frac{\partial u_1}{\partial w_{n-p}} \\ \vdots \\ \frac{\partial u_m}{\partial w_1}, \dots, \frac{\partial u_m}{\partial w_{n-p}} \end{bmatrix}$$

The differential of the constraints is

$$(47) \quad dg = G_u du + G_w dw = 0,$$

where

$$dg = [dg_1, \dots, dg_m], \quad dw = [dw_1, \dots, dw_{n-p}], \quad du = [du_1, \dots, du_m]$$

$$G_w = \begin{bmatrix} \frac{\partial g_1}{\partial w_1}, \dots, \frac{\partial g_1}{\partial w_{n-p}} \\ \vdots \\ \frac{\partial g_m}{\partial w_1}, \dots, \frac{\partial g_m}{\partial w_{n-p}} \end{bmatrix} \quad G_u = \begin{bmatrix} \frac{\partial g_1}{\partial u_1}, \dots, \frac{\partial g_1}{\partial u_m} \\ \vdots \\ \frac{\partial g_m}{\partial u_1}, \dots, \frac{\partial g_m}{\partial u_m} \end{bmatrix}$$

Since

$$(48) \quad du = U_w dw,$$

we see from Eq. (47) that

$$(49) \quad U_w = -G_u^{-1} G_w,$$

and

$$(50) \quad \Delta_c = F_w - F_u G_u^{-1} G_w.$$

The iteration formula for the independent variables can be immediately written down.

$$(51) \quad W^{p+1} = W^p - h^p B^{-1} \Delta_c^p$$

where Δ_c^p is the gradient given in Eq. (50) evaluated at W^p and U^p . To a linear approximation, the change in the dependent variables can be obtained from Eqs. (48) and (49). Thus,

$$(52) \quad U^{p+1} \approx U^p - G_u^{-1} G_w [W^{p+1} - W^p] = h^p G_u^{-1} G_w B^{-1} \Delta_c^p.$$

However, due to the linear approximation the constraints will not in general be satisfied and Eq. (47) will not hold. Since the above formulas depend on the validity of Eq. (47), the dependent variables must be corrected iteratively before the next iteration of the independent variables. In his paper Brown recommends using the Newton-Raphson method which is simply the method of steepest descent in conjunction with the scale factor given in Eq. (17). In this case, the iteration formula is:

$$(53) \quad {}_{r+1}U^{p+1} = {}_rU^{p+1} - {}_r[G_u]^{-1} {}_rG$$

where the prescript r refers to the r th iteration in the Newton-Raphson method and

$$(54) \quad {}_rG = [g_1(W^{p+1}, {}_rU^{p+1}), \dots, g_m(W^{p+1}, {}_rU^{p+1})]$$

PART III: SEQUENTIAL MIN-MAX SEARCH PROCEDURES

There exist a group of search procedures which reduce the range of the independent variables a predetermined amount at each iteration. Hence it is possible to determine, before making any tests, the number of necessary iterations to reduce the range of the independent variables a given amount. These methods work as well for the worst case in the class of functions for which they have been developed as for the best case. In that sense they are conservative and correspond to the Min-Max solutions in game theory [30]. For this reason, we have chosen to call them sequential Min-Max search procedures. Unfortunately, these methods are very difficult to obtain. For unimodal functions with one independent variable, Kiefer (30, 31) has developed a method based on the Fibonacci numbers. For more independent variables, the only method to date has been one by Newman [32]. However, this method is limited in the type of function to which it can be applied.

Throughout this section, we shall equate the number of independent variables with the dimension of the function. Thus, a function with two independent variables will be considered a two-dimensional function.

3.1. ONE-DIMENSIONAL SEARCH

The search for a minimum of a unimodal function with one independent variable has been proved optimum in a min-max sense by Kiefer [31] and independently by Johnson [33]. Let us assume that the minimum of the function initially lies within an interval a_p, b_p as shown in Fig. 8. If we pick two test points x_1^p and x_2^p we know that if

$$\begin{array}{ll} f(x_1^p) < f(x_2^p) & \text{then the minimum lies between } a_p \text{ and } x_2^p \\ f(x_1^p) > f(x_2^p) & \text{then the minimum lies between } x_1^p \text{ and } b_p \\ f(x_1^p) = f(x_2^p) & \text{then the minimum lies between } x_1^p \text{ and } x_2^p. \end{array}$$

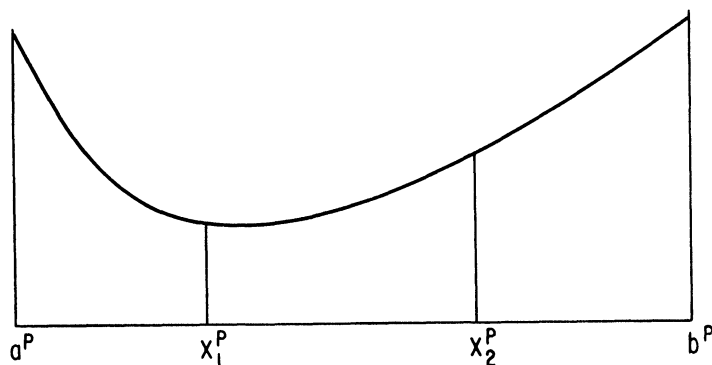


FIGURE 8. ONE DIMENSIONAL SEARCH

To maintain mathematical symmetry, we pick either the interval (a_p, x_2^p) or (x_1^p, b_p) whenever the equality condition occurs. Thus, by this simple test, we can reduce the size of the interval within which we guarantee the minimum to be located.

To obtain an optimum strategy for choosing the location of the test points, we must *a priori* determine the total number of tests, N , that will be made. We shall indicate later how N can be selected. After $p - 1$ iterations, the optimum test points are:

$$(55) \quad x_1^p = \frac{U_{N-1-p}}{U_{N+1-p}} (b_p - a_p) + a_p \quad x_2^p = \frac{U_{N-p}}{U_{N+1-p}} (b_p - a_p) + a_p$$

where $b_p - a_p$ is the length of the interval after the $p - 1$ test and U_p are the Fibonacci numbers given by the iterative equation

$$(56) \quad U_p = U_{p-1} + U_{p-2} \quad U_0 = 1, \quad U_1 = 1$$

The first few Fibonacci numbers are 1, 2, 3, 5, 8, 13, 21 and $U_{20} > 10^4$. The last test points are given by

$$(57) \quad \begin{aligned} x_1^{N-1} &= (\tfrac{1}{2} + \epsilon)(b_{N-1} - a_{N-1}) + a_{N-1} \\ x_2^{N-1} &= \tfrac{1}{2}(b_{N-1} - a_{N-1}) + a_{N-1} \end{aligned}$$

where ϵ is an arbitrarily small number. The number ϵ is introduced in this last iteration in order to determine which interval contains the minimum.

It can easily be shown that the size of the remaining interval after N iterations is

$$(58) \quad b_N - a_N = \frac{1}{2U_N} (b_0 - a_0) + \epsilon$$

where $b_0 - a_0$ is the original interval. Thus, the number of iterations for a desired accuracy can be determined from a Fibonacci table.

Due to the properties of the Fibonacci numbers, the function need only be evaluated once per iteration after the initial two tests. To show this, let us assume that at the p th iteration, the minimum is located between a_p and x_2^p . Then, at the next iteration

$$(59) \quad x_2^{p+1} = \frac{U_{N-(p+1)}}{U_{N+1-(p+1)}} (x_2^p - a_p) + a_p$$

Substituting the value of x_2^p given by Eq. (55), we see that

$$(60) \quad x_2^{p+1} = \frac{U_{N-(p+1)}}{U_{N-p}} \left(\frac{U_{N-p}}{U_{N+1-p}} (b_p - a_p) + a_p - a_p \right) + a_p$$

$$(61) \quad x_2^{p+1} = \frac{U_{N-p-1}}{U_{N+1-p}} (b_p - a_p) + a_p = x_1^p$$

A similar proof can be developed for the case of the minimum lying between x_1^p and b_p .

For large values of p , the ratios of the Fibonacci numbers given in Eq. (55) approach a constant:

$$(62) \quad \text{and} \quad \frac{U_{p-1}}{U_{p+1}} \approx 0.382$$

$$\frac{U_p}{U_{p+1}} \approx 0.618$$

Therefore, the following approximate formulas can be used to obtain x_1^p and x_2^p , ($p = 1, \dots, N$):

$$(63) \quad \begin{aligned} x_1^p &= 0.382(b_p - a_p) + a_p \\ x_2^p &= 0.618(b_p - a_p) + a_p \end{aligned}$$

After the first iteration, only one test point is calculated. It can be seen that the test points at any iteration are independent of the total number of points. After N iterations, Kiefer [31] shows that the final interval will be 13 per cent larger than the interval obtained without the approximation. However, each iteration has been simplified and the decrease in total computer time may more than compensate for the additional iteration required.

Since the size of the remaining interval after N iterations can be calculated, it is possible to determine the cost of a search. Using the above approximation, it can easily be seen that the final interval will be

$$(64) \quad \delta_N = (0.618)^N(b_0 - a_0)$$

If C_1 is the cost per test point, the total cost is

$$(65) \quad C_T = C_1(N + 1) = \frac{C_1 \log \left(\frac{\delta_N}{b_0 - a_0} \right)}{\log 0.618} + C_1$$

$$(66) \quad = -2.1C_1 \log \left(\frac{\delta_N}{b_0 - a_0} \right) + C_1$$

3.2. HIGHER DIMENSIONAL SEARCH

For more than one independent variable there is no simple point test corresponding to the test used in Kiefer's method. Newman's method [32] is based on the following theorem which is presented here without proof.

Given an n -dimensional unimodal function, $f(x_1, \dots, x_n)$ whose minimum is known to be in a given n -dimensional space and a $n - 1$ dimensional hyperplane which divides that space. If $f(X^\circ)$ is the absolute minimum of f on the hyperplane, then the negative direction of the gradient at X° determines which portion of the divided space contains the minimum.

We note that in general a unimodal function need not be unimodal on an $n - 1$ dimensional hyperplane. Thus, to use this theorem, we would need to successively make $n - 1$ dimensional multimodal searches for the absolute minimum. This is clearly not practical. However, if the unimodal function is also

unimodal on any lower dimensional projection, the search can be greatly simplified.

In the following discussion, we shall restrict ourselves to a two-dimensional function. It will become clear that the search procedure can be easily extended to higher dimensions. As shown in Fig. 9, let the minimum be contained in an area, A , and the center of gravity of A be at point X . We can now draw a line, BC , through X which is perpendicular to the gradient at X . By this construction, the absolute minimum of the function along BC occurs at X . Thus, the line BC divides the area, A , and by the above theorem, the direction of decrease of the gradient will determine which portion contains the minimum. Repeating this procedure, we can reduce the area within which we can guarantee the minimum to any desired size.

Newman [32] has shown that by using the center of gravity, the remaining area is always less than $\frac{5}{8}$ the original area. Using this fact, it is possible to show that the cost for two-dimensions is

$$(67) \quad C_T \leq 12.5C_1 \log \left(\frac{\delta_N}{b_0 - a_0} \right) + 5C_1$$

where δ_N is the final uncertainty interval for each variable, $b_0 - a_0$ is the original interval for each variable and C_1 is the cost per test point. For n dimensions the cost is

$$(68) \quad C_T \leq 3.2C_1 n^2 \left[\log \left(\frac{\delta_N}{b_0 - a_0} \right) + \log n + 1 \right]$$

where the constants are the same as defined above.

We should note that the requirement that the function be unimodal on all projections restricts the type of functions to which this method can be applied considerably, especially for high dimensions. The other difficulty is finding the center of gravity at each iteration. After a few iterations, the area will no longer

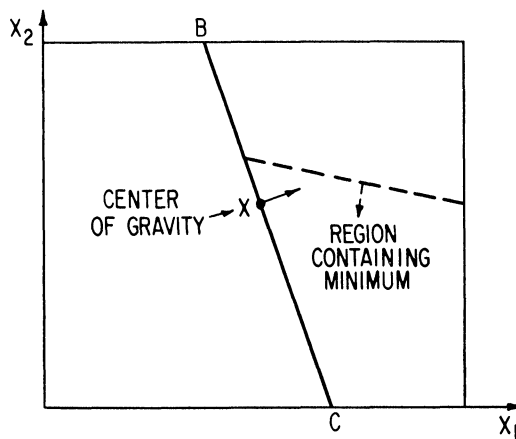


FIGURE 9. NEWMAN'S METHOD

have any symmetrical properties. Thus, the computer storage and the calculation procedures become more difficult and time consuming.

PART IV: RANDOM METHODS

The random methods are nonsequential in that the previous test points do not determine exactly where the next test point will be located. Rather, the independent variables are chosen at random according to an n -dimensional probability density function. The shape of the density function depends on the *a priori* knowledge of the programmer and is generally chosen as flat such that a particular test point in a region is as likely to occur as any other.

In the purely random method [34], the test points are picked at random in the area where the minimum is located according to a fixed distribution. After a certain number of tests have been made, the smallest value of the function is assumed to be the minimum. How many test points are necessary, can be determined in a probabilistic sense.

Let us assume that the minimum is equally likely to be anywhere in an n -dimensional hypercube with all sides equal to d . It is then reasonable to use a flat density function over the interval 0 to d . Let us further assume that we wish to reduce the uncertainty of each independent variable to δ_N . We know that the ratio of this smaller hypercube to the original hypercube is

$$(69) \quad a = \left(\frac{\delta_N}{d}\right)^n$$

The probability that a test point is *not* in the smaller hypercube is $(1 - a)$ and the probability that p trials will fail is $(1 - a)^p$. Thus, the probability of at least one point being in that hypercube is

$$(70) \quad S = 1 - (1 - a)^p$$

S can also be considered the confidence level that one of the test points will be in the hypercube. Solving for p , we see that the required number of points is

$$(71) \quad p = \frac{\log(1 - S)}{\log(1 - a)}$$

In Table I, the values of p are given for various values of a and S . It can be seen

TABLE I
THE EFFECT OF a AND S ON THE NUMBER OF TEST POINTS [FROM BROOKS (34)]

a	S			
	.8	.9	.95	.99
.1	16	22	29	44
.05	32	45	59	90
.025	64	91	119	182
.01	161	230	299	459
.005	322	460	598	919

that the required number of points increases quite rapidly with a reduction in a .

Brooks [34] asserts that the number of necessary tests points in the random method is independent of the number of independent variables. As we can see from Eqs. (69) and (70), this means that a must be a constant which requires that the ratio of δ/d must increase as n increases. Thus, the accuracy per independent variable is decreased. To see the effect of dimension on the number of tests for constant accuracy per variable, we can substitute Eq. (69) into Eq. (70).

$$(72) \quad p = \frac{\log (1 - S)}{\log \left[1 - \left(\frac{\delta_N}{d} \right)^n \right]}$$

Since $\left(\frac{\delta_N}{d} \right)^n$ will be very small, we can approximate the denominator by the first few terms of its series expansion.

$$(73) \quad p \approx - \left(\frac{d}{\delta_N} \right)^n \log (1 - S)$$

For a confidence level, S , of 0.9

$$(74) \quad p \approx 2.3 \left(\frac{d}{\delta_N} \right)^n$$

We see that the number of points goes up approximately exponentially with the number of dimensions.

It is instructive to compare this number of tests points which are necessary with a simple grid test. In the grid test, the points are equally spaced a distance δ_N apart through the space as shown in Fig. 10. All points are tested and the lowest point is considered the minimum. It is clear that in this case we would

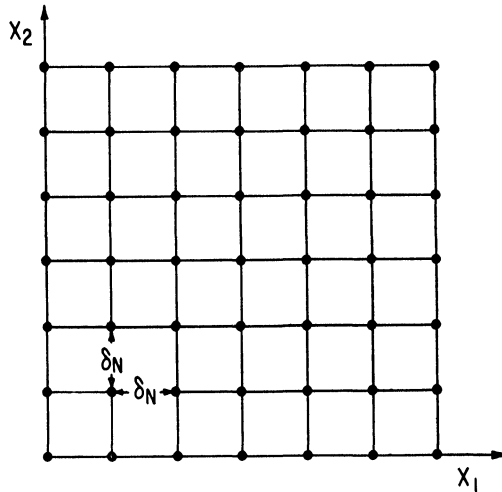


FIGURE 10. A GRID SEARCH

have to test

$$(75) \quad p = \left(\frac{d}{\delta_N} \right)^n$$

points which is approximately half the number necessary for the purely random method. It still increases exponentially and is, therefore, not a practical method for high dimensions.

In an attempt to reduce the number of necessary points, a quasisquential method [35] has been developed. It is assumed that the minimum will be located in an area around the lowest set of test points. After a certain number of tests have been made, a reduced area is selected around the lowest points. Another random search is made in this area and the process is continued. The number of points picked at random per iteration need not be as large as in the purely random method. However, it is no longer possible to obtain an estimate of the number of necessary test points or the best number of iterations since the probabilities now depend on the particular function which is being minimized.

PART V: CONCLUSION

In the preceding parts of this paper, we have discussed in detail some of the more significant search techniques. It would be useful to be able to compare the various methods so that one could select a particular method for a given function. This is generally a very difficult, if not impossible, problem. By trying each method on a set of functions such as was done by Brooks [4], one can arrive at an average statistical performance criterion. How useful such a criterion would be is debatable since there will always be some particular function for which a given method is best suited.

We can, however, point out certain fundamental characteristics of the methods which affect their performance. It is clear that Newton's and the steepest descent methods are best used on functions which satisfy the following criteria:

1. The gradient of the function is analytically available.
2. The function around the minimum depends primarily on its second derivatives.
3. The function is unimodal.

If the gradient of the function cannot be determined analytically, a finite difference approximation to the gradient must be made. In that case it would appear that the "Direct Search" procedure would be preferable. The "Direct Search" may also be useful in those cases which have complex constraints since they can be handled very easily.

The sequential min-max techniques are most desirable on peculiarly shaped functions and in those applications in which there is a high premium for a known rate of convergence. It is evident that the gradient methods will be considerably faster than these methods for functions which are nearly quadratic. Unfortunately, except for functions of one independent variable, the methods have not been developed sufficiently to reach their full potential.

The primary use for random methods is in those cases where it is not possible to do the minimization sequentially. They can also be used to determine whether

a function is multimodal and to obtain a rough idea of the nature of the surface. From this, the initial points for sequential tests can be selected. These random methods are also useful in those cases where the exact value of the function to be minimized cannot be obtained due to noise.

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